The L-Sieve, Triangular Stabilization, and the Jumps of A214972

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Abstract

We prove the conjectured jump description of A214972, the proposed recurrence for A152009, and deterministic convergence of iterated L-sieves to the triangular numbers. For any nonempty deletion set with infinite complement, the first k coordinates stabilize after at most $2k-1$ transforms. No probabilistic assumption on the initial set is needed.

1 The L-sieve as iteration of an increasing map

For a nonempty set $A \subset \mathbb{N}$ with infinite complement, the L-sieve starts with the positive integers. It records and removes the first surviving term, deletes the terms whose positions in the remaining list belong to A, and repeats. Let $S(A)$ denote the increasing sequence of recorded terms. This is the transform defined in A152009 [1]. Write $b_1 < b_2 < \cdots$ for the complement of A and put $F_A(j) = 1 + b_j$.

Lemma 1. *The r-th recorded term is $S(A)_r = F_A^{r-1}(1)$. Moreover $S(A)$ is an infinite nonempty set with infinite complement, so its L-sieve can be iterated indefinitely.*

Proof. If the current survivor list is $L(1), L(2), \ldots$, removing its first term and then deleting positions in A leaves $L(1+b_1), L(1+b_2), \ldots$. Starting from $L(j) = j$, the successive survivor maps are the iterates of F_A; the recorded terms are their values at 1. Since $b_j \geq j$, we have $F_A(j) \geq j+1$, and the orbit is infinite. As A is nonempty, $b_j \geq j+1$ for all sufficiently large j. The recorded terms consequently have eventual gaps at least 2, proving that their complement is infinite. □

2 Triangular convergence with a uniform stabilization bound

Put $T_j = j(j+1)/2$.

Theorem 1. *For every nonempty $A \subset \mathbb{N}$ with infinite complement, every $k \geq 1$, and every $r \geq 2k-1$, the first k terms of $S^r(A)$ are $T_1, \ldots, T_k$. In particular $S^r(A)$ converges coordinatewise to the triangular numbers, which form a fixed point of S.*

Proof. Suppose an increasing set A begins with $T_1, \ldots, T_k$. For $j < k$, exactly j deleted positions lie at most $T_{j+1} - 1 = T_j + j$, and the last of these positions is smaller than that endpoint. Thus $b_{T_j} = T_{j+1} - 1$, or $F_A(T_j) = T_{j+1}$. For $j = k$, at least k deleted positions lie below the T_k-th retained position, so $F_A(T_k) \geq T_{k+1}$. Lemma 1 now shows that $S(A)$ begins with $T_1, \ldots, T_k$ and that its next term is at least T_{k+1}. In this new deletion set exactly k positions lie at most $T_{k+1} - 1$, so one more application gives $S^2(A)_{k+1} = T_{k+1}$.

Every first transform starts with $T_1 = 1$. Applying the preceding argument inductively extends the stabilized prefix by one coordinate every two transforms. A single transform preserves any already triangular prefix, which gives the assertion for every later r, not just odd r. The same complement calculation for all j proves that the entire triangular sequence is a fixed point. □

The entry phrases convergence for a random positive sequence. The theorem provides a deterministic version on the natural domain of infinite iteration. The domain condition matters: deleting every positive position leaves no second recorded term, so such inputs cannot support an infinite transform. Repeated or unordered deletion indices, when allowed, are interpreted as a set.

3 Exact recurrence and all jumps

For an integer $d \geq 2$, let $A_d = \{1, 1+d, 1+2d, \ldots\}$ and write $s_r^{(d)} = S(A_d)_r$. The j-th retained position is $b_j = j + \lceil j/(d-1) \rceil = \lceil dj/(d-1) \rceil$. Lemma 1 gives the exact recurrence

$$s_1^{(d)} = 1, \qquad s_{r+1}^{(d)} = 1 + \left\lceil \frac{d\, s_r^{(d)}}{d-1} \right\rceil. \tag{1}$$

In particular, for $d = 3$ this is the proposed A152009 formula

$$s_1 = 1, \qquad s_{r+1} = \left\lfloor \frac{3s_r + 3}{2} \right\rfloor.$$

Define $h_d(0) = 0$ and, for $n \geq 1$,

$$h_d(n) = 1 + h_d\left(\left\lfloor \frac{(d-1)(n-1)}{d} \right\rfloor \right).$$

The argument on the right is nonnegative and smaller than n.

Theorem 2. *For every $n \geq 0$,*

$$h_d(n) = \#\{r \geq 1 : s_r^{(d)} \leq n\}.$$

Consequently $h_d(n) - h_d(n-1)$ is 1 exactly at the L-sieve values $s_r^{(d)}$, and is zero elsewhere. The case $d = 3$ proves the conjecture in A214972 [2].

Proof. Induct on r to prove $h_d(n) \geq r$ if and only if $n \geq s_r^{(d)}$. For $r = 1$ both conditions mean $n \geq 1$. For $r+1$, the recurrence and induction give

$$h_d(n) \geq r+1 \iff \left\lfloor \frac{(d-1)(n-1)}{d} \right\rfloor \geq s_r^{(d)} \iff n \geq 1 + \left\lceil \frac{d s_r^{(d)}}{d-1} \right\rceil.$$

The last threshold is (1). The thresholds strictly increase, so counting them yields the claimed formula and jump description. □

4 Independent checks

The accompanying standard-library verifier compares direct recursion with threshold counting, reconstructs L-sieve terms by literal deletion, checks the published A152009 values, and tests triangular stabilization on exhaustive finite deletion sets and several infinite periodic sets. These finite controls check the implementation; the theorems prove the unbounded statements. The recurrence and conjectures are credited to the OEIS entries; no claim of an exhaustive literature-priority search or OEIS acceptance is made.

References

[1] J. W. Layman, OEIS A152009 (2008), https://oeis.org/A152009, accessed October 26, 2026.

[2] C. Kimberling, OEIS A214972 (2012), https://oeis.org/A214972, accessed October 26, 2026.